

SURVEYING

- The art and science of determining angular and linear measurements to establish the form, extent, and relative position of points, lines, and areas on or near the surface of the earth or on other extraterrestrial bodies through applied mathematics and the use of specialized equipment and techniques.

CLASSIFICATION OF SURVEYING

1. PLANE SURVEYING

- a type of surveying where the earth is considered as a flat surface, and where distances and areas involved are of limited extent

2. GEODETIC SURVEYING

- surveys of wide extent which take into account the spheroidal shape of the earth

TYPES OF SURVEYS

1. CADASTRAL SURVEYS

- closed surveys in urban and rural locations to determine and define property lines and boundaries, corners, and areas.

2. CITY SURVEYS

- surveys in the city for planning expansions, locating property lines, fixing reference monuments, determining physical features of land, and preparing maps.

3. CONSTRUCTION SURVEYS

- surveys done at a construction site to provide data regarding grades, reference lines, dimensions, ground configuration, and location and elevation of structures.

4. FORESTRY SURVEYS

- survey in connection with forest management and mensuration, and the production and conservation of forest lands.

5. HYDROGRAPHIC SURVEYS

- surveys made to map shore lines, chart the shape of areas underlying water surfaces, and measure the flow of streams.

6. INDUSTRIAL SURVEYS

- known as optical tooling
- surveys for ship building, construction and assembly of aircraft, layout and installation of heavy and complex machinery, and for industries requiring very accurate dimensional layouts.

7. MINE SURVEYS

- surveys to determine the position of all underground excavations and surface mine structures, to fix surface boundaries of mining claims, determine geological formations, calculate excavated volumes, and establish lines and grades for other related mining work.

8. PHOTOGRAMMETRIC SURVEYS

- survey which uses photographs taken with specially designed cameras either from airplanes or ground stations.

9. ROUTE SURVEYS

- determination of alignment, grades, earthwork quantities, location of natural and artificial objects in connection with the planning, design, and construction of highways, railroads, pipelines, canals, transmission lines, and other linear projects.

10. TOPOGRAPHIC SURVEYS

- surveys to determine the shape of the ground, the location and elevation of natural and artificial features upon it.

DEVELOPMENT OF SURVEYING INSTRUMENTS

- The extensive use of surveying instruments came during the early days of the Roman Empire.

1. ASTROLABE

- invented by Hipparchus in 140 BC and further improved by Ptolemy
- used to determine the altitude of stars

2. TELESCOPE

- invented by Lippershey (1607)
- Galileo constructed a refracting telescope for astronomical observations in 1609.
- Only used in surveying after the cross hairs for fixing the line of sight were introduced

3. TRANSIT

- invented by Young and Draper (1830)
- the universal surveying instrument

4. SEMICIRCUMFERENTOR

- used to measure and lay off angles and establish lines of sight by employing peep sights

5. PLANE TABLE

- used in field mapping
- consists of a board attached to a tripod and can be leveled or rotated to any direction

6. DIOPTRA

- invented by Heron of Alexandria
- used in leveling and measuring horizontal and vertical angles

7. ROMAN GROMA

- for aligning or sighting points
- consists of cross arms with suspended plumb lines fixed at right angles and pivoted upon a vertical staff.

8. LIBELLA

- used by Assyrians and Egyptians
- had an A-frame with a plumb line suspended from its apex
- used to determine the horizontal

9. VERNIER

- invented by Pierre Vernier
- a short auxiliary scale placed alongside the graduated scale of an instrument to determine the fractional parts of the main scale without interpolating.

10. DIOPTER

- developed by Greeks in 130 BC
- used for leveling, laying off right angles, and measuring horizontal and vertical angles.

11. COMPASS

- for determining direction of lines and calculating angles between lines
- consists of a magnetized steel needle which points at the magnetic north

12. GUNTER'S CHAIN

- invented by Sir Edmund Gunter (1620)
- used for taping distances
- 66 ft long and contains 100 links

13. CHOROBATES

- for leveling work
- consists of a horizontal straight-edge about 6 m long, a groove 2.5 cm deep and 1.5 m long on top.
- water is poured into the groove for leveling

14. MERCHET

- for measuring time and meridian
- used by Chaldeans in 4000 BC
- consists of a slotted palm leaf through which to sight and a bracket from which a plumb bob was suspended

MEASUREMENT

- process of determining the extent, size, or dimension of a particular quantity in comparison to a given standard

TYPES OF MEASUREMENTS

1. DIRECT MEASUREMENTS

- comparison of the measured quantity with a standard measuring unit (makes use of instruments to measure the quantity)

2. INDIRECT MEASUREMENTS

- made when it is not possible to apply a measuring instrument directly to a quantity

THE METER

- proposed in 1789 by French scientists to establish a system suitable for all times and all peoples, and which could be based upon permanent natural standards
- originally defined as 1/10,000,000 of the earth's meridional quadrant
- a treaty was signed in Paris in May 20, 1875 by representatives of 18 countries creating the International Bureau of Weights and Measures.
- the treaty established the International Meter which is based on an iron meter bar standardized in Paris in 1799
- defined as the distance between two lines engraved across the surface of a bar with an X-shaped cross section, composed of 90% platinum and 10% iridium when the temperature of the bar is 0°C.
- it was redefined in 1960 as the length equal to 1,650,763.73 wavelengths of the orange-red light produced by burning the element Krypton (atomic number 86) at a specified energy level in the spectrum.
- Effective January 1, 1983, the English System was officially phased out in the Philippines and only the modern metric system was allowed to be used. The metric conversion was signed into law in 1978 by Pres. Marcos. The metric system prescribed the use of the International System of Units (SI) as established by the General Conference on Weights and Measures and as modified by the local Metric System Board to suit Philippine Conditions.

UNITS OF MEASUREMENT

1. Linear, Area, and Volume Measurements

2. Angular Measurements

- Radian – SI unit for plane angles (the angle subtended by an arc of a circle having a length equal to the radius of a circle)

a. Sexagesimal Units

- the degree, minute, second

b. Centesimal Units

- uses grads ($360^\circ = 400$ grads)
- the grad is divided into 100 centesimal minutes and the minute is divided into 100 centesimal seconds

SURVEYING FIELD NOTES

- constitute the only reliable and permanent record of actual work done in the field

FIELD NOTEBOOK

- used to record the field notes

TYPES OF NOTES

1. Sketches
2. Tabulations
3. Explanatory Notes
4. Computations
5. Combination of the above

INFORMATION FOUND IN FIELD NOTEBOOKS

1. Title of the Field Work or Name of Project
2. Time of Day and Date
3. Weather Conditions
4. Names of Group Members and Their Designations
5. List of Equipment

THE FIELD SURVEY PARTY

1. **Chief of Party**
 - responsible for the overall direction, supervision, and operational control of the survey party.
2. **Assistant Chief of Party**
 - assists the chief of party in the accomplishment of the task assigned to the survey party.
3. **Instrument Man**
 - sets up, levels, and operates the surveying instruments.
4. **Technician**
 - responsible for the use and operation of all electronic instruments required in the field.
5. **Computer**
 - performs all computations of survey data and works out necessary computational checks required in the field.

6. Recorder

- keeps the record of all sketches, drawings, measurements, and observations taken in the field.

7. Head Tapeman

- responsible for the accuracy and speed of all linear measurements with tape.
- carries the zero end of the tape ahead

8. Rear Tapeman

- assists the head tapeman during taping operations and in other related work.
- holds the 30-m end or any intermediate meter mark of the tape during measurement

9. Flagman

- holds the flagpole or range pole at selected points as directed by the instrument man.

10. Rodman

- holds the stadia or leveling rod when sights are to be taken on it.

11. Pacer

- checks all linear measurements made by the tapeman. May also perform the job of the rodman.

12. Axeman/Lineman

- clears the line of sight of trees, brush, and other obstructions in wooded country.

13. Aidman

- renders first aid treatment to members of the survey party who are involved in snake or insect bites, accidents and other health issues. May also be designated as assistant instrument man.

14. Utilitymen

- renders other forms of assistance needed by the survey party or as directed by the chief of party.

ERROR

- difference between the true value and the measured value of a quantity

MISTAKES

- inaccuracies in measurements which occur because some aspect of a surveying operation is performed by the surveyor with carelessness, inattention, poor judgment, and improper execution.
- large mistake → blunder
- not classified as errors (so large in magnitude)

TYPES OF ERRORS

1. SYSTEMATIC ERRORS

- one which will always have the same sign and magnitude as long as field conditions remain constant and unchanged.

2. ACCIDENTAL ERRORS

- the occurrence of such errors are matters of chance as they are likely to be positive or negative, and may tend in part to compensate or average out according to laws of probability.

SOURCES OF ERRORS

1. INSTRUMENTAL ERRORS

- due to imperfections of the instrument used, either from faults in their construction or from improper adjustments between the different parts prior to their use.

2. NATURAL ERRORS

- caused by variations in the phenomena of nature such as changes in magnetic declination, temperature, humidity, wind, refraction, gravity, and curvature of the earth.

3. PERSONAL ERRORS

- arise from the limitations of the senses of sight, touch and hearing of the human observer which are likely to be erroneous or inaccurate.

ACCURACY

- how close a given measurement is to the absolute or true value of the quantity measured.

PRECISION

- degree of refinement and consistency with which any physical measurement is made

PROBABILITY

- number of times something will probably occur over the range of possible occurrences.
- in dealing with probability, it is assumed that we only refer to accidental errors and that all systematic errors and mistakes are eliminated.

THEORY OF PROBABILITY

1. small errors occur more often than large ones and that they are more probable
2. large errors happen infrequently and are less probable

3. positive and negative errors of the same size happen with equal frequency, and they are equally probable
4. the mean of an infinite number of observations is the most probable value

MOST PROBABLE VALUE

- refers to a quantity which, based on available data, has more chance of being correct than has any other

$$MPV = \bar{X} = \frac{\sum X}{n} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n}$$

RESIDUAL (DEVIATION)

- difference between any measured value of a quantity and its most probable value

$$v = X - \bar{X}$$

- residuals and errors are theoretically identical but errors cannot be calculated because there is no way of knowing true values.

PROBABLE ERROR

- quantity which, when added to or subtracted from the most probable value, defines a range within which there is a 50% chance that the true value of the measured quantity lies inside (or outside) the limits thus set.

Probable Error of any Single Measurement of a Series:

$$PE_s = \pm 0.6745 \sqrt{\frac{\sum v^2}{n-1}}$$

Probable Error of the Mean:

$$PE_m = \pm 0.6745 \sqrt{\frac{\sum v^2}{n(n-1)}}$$

RELATIVE ERROR (RELATIVE PRECISION)

- ratio of the error to the measured quantity

INTERRELATIONSHIP OF ERRORS

$$Weight = \frac{1}{PE^2}$$

Summation of Errors:

$$PE_s = \pm \sqrt{PE_1^2 + PE_2^2 + PE_3^2 + \dots + PE_n^2}$$

Product of Errors:

$$PE_p = \pm \sqrt{(Q_1 \times PE_2)^2 + (Q_2 \times PE_1)^2}$$

MEASUREMENT OF HORIZONTAL DISTANCES

1. BY PACING

- Pacing – counting the number of steps or paces in a required distance
- Pace – length of a step in walking
- Stride – double step (1 stride = 2 paces)
- Relative Precision: 1/200
- Pace Factor – in meters/pace
- **Factors Affecting Length of Pace:**
 - o Speed of pacing
 - o Roughness of the ground
 - o Weight of clothing and shoes used
 - o Fatigue on part of the pacer
 - o Slope of the terrain
 - o Age and sex of the individual
- **Mechanical Pace Counters:**
 - o Pedometer
 - records the number of steps made on the body's movement
 - o Passometer
 - strapped to the leg of the pacer
 - registers a pace by impact each time a foot touches the ground

2. BY TAPING

- the use of a graduated tape is the most common method of measuring horizontal distances
- Chaining – measurement of distances using chains
- Chainmen – persons undertaking measurement using chains

3. BY TACHYMETRY

- also called Tacheometry
- based on the optical geometry of the instruments employed (indirect method of measurement)
- **a. Stadia Method**
 - introduced by James Watt (1771)
 - relative precision: 1/300 to 1/1000
 - consists of a telescope with two horizontal hairs (stadia hairs) and a graduated rod (stadia rod)
 - factors affecting precision:
 - o refinement with which the instrument was manufactured
 - o skill of the observer
 - o length of measurement
 - o effects of refraction and parallax

$$D = Ks + C$$

where:

- K** – stadia interval factor of the instrument
- s** – difference between the upper stadia hair reading and lower stadia hair reading
- C** – distance from the center of the instrument to the principal focus (instrument constant)

b. Subtense Bar Method

- bar: 2 meters long
- horizontal distance is measured by setting up the subtense bar at the distant station and measuring the horizontal angle subtended by the distance between the two targets using a theodolite.
- horizontal distance is obtained directly and no slope correction is required
- relative precision: 1/3000 for sights of 150 m or less and using a 1 sec theodolite.

$$D = \frac{s}{2} \cot\left(\frac{\alpha}{2}\right)$$

where:

- s** – length of the subtense bar
- α** – angle subtended by the targets

4. BY GRAPHICAL AND MATHEMATICAL METHODS

- Unknown distances may be determined through their relationship with known distances geometrically.

5. BY MECHANICAL DEVICES

a. Odometer

- a device attached to a wheel for measuring surface distances
- precision: 1/200 (on fairly level terrain)

b. Measuring Wheel

- a more portable and self-contained measuring device than the odometer
- consists of a small wheel attached to a rod and handle

c. Optical Rangefinder

- usually handheld or mounted on a small tripod
- determines distances by focusing
- precision: 1/50 (for distances less than 500 meters)

6. BY PHOTOGRAMMETRY

- measurement of images on a photograph
- precision: 1/3000 to 1/5000

ELECTRONIC DISTANCE MEASUREMENT (EDM)

- based on the basic principle that the time required for a radio or light wave to travel from one end of a line to the other is a function of the length measured.
- **Classification:**
 - a. **Electro-Optical Instruments**
 - transmit wavelengths within or slightly beyond the visible region of the spectrum
 - b. **Microwave Instruments**
 - transmits microwaves with wavelengths of 1.0 to 8.6 mm
 - consists of two identical units which includes a transmitter and a receiver
- **Advantages:**
 - speed and accuracy in measurement
 - lightweight and portable
 - low power requirement
 - easy to operate
 - ease in measurement over inaccessible terrain
 - applicable to the measurement of short and long lengths
 - automatically measures displays and records, slope range, azimuth, vertical angle, horizontal distance, departure and latitude
 - slope measurements are internally reduced to horizontal and vertical components by built-in computers thereby eliminating the need to calculate these values
 - automatically accumulates, and averages reading for slope range, horizontal and vertical angles
- **Operating Ranges:**
 - Short range – distances less than 25 km
 - Medium range – distances from 25 km to 75 km
 - Long range – distances of 75 km or longer

GEODIMETER

- acronym for geodetic distance meter
- an electro-optical device developed by Erik Bergstrand (1948)
- maximum range (5 to 10 km during daytime, 25 to 30 km during nighttime)
- precision: 1/200,000

TELLUROMETER

- world's second EDM instrument
- uses high frequency microwave transmission capable of measuring distances up to 80 km day or night
- precision: 1/300,000

MEASURING TAPES

1. Steel Tape

- also known as surveyor's or engineer's tape
- made of a ribbon of steel 0.5 to 1.0 cm in width
- weights 0.8 to 1.5 kg per 30 meters
- for most conventional measurements in surveying and engineering work

2. Metallic Tape

- also called woven tapes
- made of water-proof linen fabric with woven small brass, copper, or bronze wires to increase its strength and reduce stretching
- for measuring short distances

3. Non-metallic Tape

- woven from synthetic materials with strong dimensional stability
- coated with a plastic material to reduce effects of moisture, humidity, and abrasion

4. Invar Tape

- made of an alloy of nickel (35%) and steel (65%)
- very low coefficient of thermal expansion (1/30 to 1/60 that of a steel tape)
- "invariable" → less affected by temperature changes than steel tape
- used for precise measurements
- 10 times as expensive as ordinary steel tapes

5. Lovar Tape

- properties and costs between conventional steel tapes and invar tapes

6. Fiberglass Tape

- woven with fiberglass in a longitudinal and transverse pattern
- does not shrink or stretch with changes in temperature and humidity
- best used in vicinity of electrical equipment

7. Wires

- used before thin flat steel tapes were produced
- still used in hydrographic survey

8. Builder's Tape

- have smaller cross sections and lighter than steel tape
- used in building construction

9. Phosphor-Bronze Tape

- rust-proof tape to use in the vicinity of salt water

10. Nylon-coated Steel Tape

- coated with permanently bonded non-conducting nylon
- resistant to corrosion and immune to rust

TAPING ACCESSORIES

1. Range Pole

- also known as flags or lining rods
- used as temporary signals to indicate the location of points or direction of lines, and to mark alignment.
- usual length: 2.0 or 3.0 meters
- marked with alternate red and white sections 30 cm or 50 cm long

2. Tape Clamping Handles

- applies tension with a quick grip on any part of a steel tape without causing damage to the tape or hands of the tapeman

3. Chaining Pins

- also known as surveyor's arrows or taping pins
- stuck in the ground to mark the ends of a measured tape lengths or partial tape lengths
- made of heavy wire (30 cm long) and painted with alternate red and white bands
- sets of 11 pins carried on a steel ring are standard

4. Tension Handle

- also known as a spring scale
- used at one end of a tape for ensuring the application of the correct amount of pull on the tape during measurement
- used in precision taping

5. Tape Thermometer

- used to determine the temperature of the air and the approximate temperature of the tape during measurement
- about 10 to 15 cm long and is graduated from -30 to 50°C in 2 or 5 degree divisions

6. Plumb Bob

- used for projecting the tape ends to the ground when the tape must be suspended above the measured line

- weights 0.25 kg and attached to 1.5 m long string or cord which is free of knots

7. Wooden Stake or Hub

- made of 5 cm x 5 cm x 30 cm wood to mark points, corners, or stations in the ground

8. Leather Thongs

- attached to a ring located near the zero-meter mark of the tape to provide a comfortable grip on the tape when measuring

9. Hand Level and Clinometer

- a 15 cm long device which consists of a metal sighting tube with a level bubble
- used to keep the tape ends at equal elevations when measuring over rugged terrain, in approximately determining difference in elevation of points, and in other field operation where it is required to produce a level sight

10. Tape Repair Kit

- allows emergency repairs to be made on damaged or broken tapes

11. Crayons

- marking crayons used in surveying are usually lumber crayons
- about 10 cm long and hexagonal in cross section (usually in blue, yellow, and red color)
- used for marking points, corners, or stations by indicating cross marks on paved roads, sidewalks or walls

PROCEDURE OF TAPING

1. Aligning the tape
2. Stretching the tape
3. Plumbing
4. Marking full tape lengths
5. Tallying taped measurements
6. Measuring fractional lengths

* If the tape is stretched less than the standard pull, all recorded measurements will turn out to be "too short". If stretched greater than the standard pull, all measurements will be "too long".

BREAKING TAPE

- measurement of shorter distances which are accumulated to total a full tape length

SLOPE TAPING

- tape measurements made directly along the slopes when the ground is of uniform inclination and fairly smooth

$$d = s \cos \alpha$$

$$d = \sqrt{s^2 - h^2}$$

where:

d – horizontal distance

s – slope distance

α – angle of inclination of the slope

h – difference in elevation

CORRECTIONS IN TAPING

	Too Long	Too Short
Measuring	+	-
Laying Out	-	+

a. Correction due to Incorrect Tape Length

$$Corr = TL - NL$$

$$C_1 = Corr \left(\frac{ML}{NL} \right)$$

$$CL = ML \pm C_1$$

where:

TL – true or actual length of the tape

NL – nominal length of the tape

ML – measured length

C_1 – total correction to be applied

CL – corrected length

b. Correction due to Slope

$$d = s - C_h$$

- **Gentle Slopes** (Less than 20%)

$$C_h = \frac{h^2}{2s}$$

- **Steep Slopes** (20% to 30%)

$$C_h = \frac{h^2}{2s} + \frac{h^4}{8s^3}$$

- **Very Steep Slopes** (Greater than 30%)

$$C_h = s(1 - \cos \theta)$$

c. Correction due to Alignment

- the linear error due to inaccuracy in alignment of a tape is similar to the effect of slope and can be computed in the same manner

d. Correction due to Temperature

$$C_T = \alpha L(T - T_o)$$

where:

α – coefficient of linear expansion per degree change in temperature

L – length of the tape measured

T – observed temperature

T_o – temperature of standardized tape

For steel tapes: **$\alpha = 0.0000116/^{\circ}\text{C}$**

e. Correction due to Tension

$$C_P = \frac{(P - P_o)L}{AE}$$

$$A = \frac{W}{L\gamma}$$

where:

P – measured pull

P_o – standard pull

L – length of tape measured

A – cross sectional area of the tape

E – modulus of elasticity of the tape

W – total weight of the tape

γ – unit weight of the tape

For steel tapes:

$$\gamma = 7.866 \times 10^{-3} \text{ kg/cm}^3$$

$$A = 0.02 \text{ to } 0.06 \text{ cm}^2$$

$$E = 2.00 \times 10^6 \text{ to } 2.10 \times 10^6 \text{ kg/cm}^2$$

f. Correction due to Sag

- sag shortens the horizontal distance between end graduations of the tape

$$C_s = \frac{w^2 L^3}{24P^2}$$

$$C_s = \frac{W^2 L}{24P^2}$$

$$W^2 = w^2 L^2$$

where:

w – weight of tape per unit length

W – total weight of tape between supports

L – interval between supports (unsupported length of tape)

P – pull or tension applied on the tape

g. Correction due to Wind

- its effect is similar to the effect of sag but usually much less

h. Normal Tension

- the applied pull lengthens the tape to equal the shortening caused by sag

$$P_N = \frac{0.204 W \sqrt{AE}}{\sqrt{P_N - P_S}}$$

where:

P_N – normal tension or pull to eliminate the effect of sag

P_S – standard pull for the tape

W – total weight of tape between supports

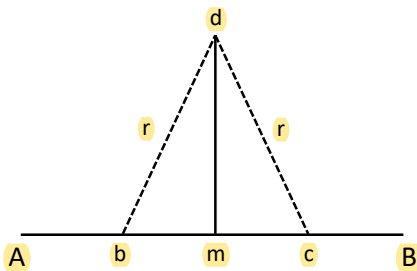
A – cross sectional area of tape

E – modulus of elasticity of tape

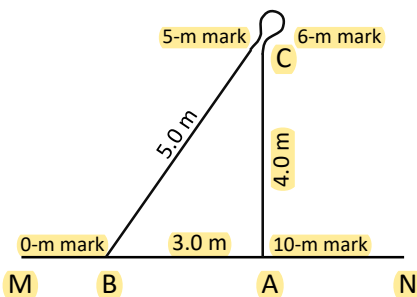
SURVEYS WITH TAPE

1. Erecting Perpendicular to Line

a. Chord-Bisection Method



b. 3:4:5 Method



2. Measuring Angles with Tape

3. Laying Off Angles with Tape

4. Determining Obstructed Distances

LEVELING

- process of directly or indirectly measuring vertical distances to determine the elevation of points or differences in elevation.

DEFINITION OF TERMS

1. Level Surface

- a curved surface which is at any point perpendicular to the direction of gravity or the plumb line

2. Level Line

- a curved line in a level surface all points of which are normal to the direction of gravity and equidistant from the center of the earth

3. Horizontal Surface

- plane that is tangent to a level surface at a particular point

4. Horizontal Line

- a straight line in a horizontal plane which is tangent to a level line at one point

5. Vertical Line

- a line parallel to the direction of gravity

6. Mean Sea Level

- an imaginary surface of the sea which is midway between high and low tides

7. Vertical Datum

- any convenient level surface coincident or parallel with mean sea level to which elevations of a particular area are referred

8. Elevation

- vertical distance above or below sea level or any other selected datum

9. Difference in Elevation

- vertical distance between the two level surfaces in which the points lie

LEVELING METHODS

1. Direct or Spirit Leveling

- the most commonly employed method of determining the elevation of points some distance apart by a series of set ups of a leveling instrument along a selected route
- uses a spirit level

2. Reciprocal Leveling

- process of accurately determining the difference in elevation between two intervisible points located at a considerable distance apart

3. Profile Leveling

- used to determine differences in elevation between points at designated short measured intervals along an established line to provide data from which a vertical section of the ground surface can be plotted

4. Trigonometric Leveling

- used to determine the difference in elevation between two points by trigonometric computations from measurements of its

horizontal or slope distance and the vertical angle between the points

5. Stadia Leveling

- combines features of direct leveling with those of trigonometric leveling
- differences in elevation between points are computed from observed vertical angles and the three intercepts on a rod held at each point backsighted or foresighted

6. Barometric Leveling

- determination of differences in elevation between points by measuring the variation in atmospheric pressure at each point by means of a barometer

7. Cross-Section Leveling

- used to obtain a representation of the ground surface on either side of the centerline

8. Borrow-Pit Leveling

- method of determining the relative elevations of points in borrow-pit excavations for the purpose of calculating volumes of earthwork

TYPES OF LEVELS

1. Dumpy Level

- has a long telescope attached to the level bar
- the telescope, which can be rotated 360°, fixes the direction of the line of sight
- attached to the level bar is the level vial which always remain in the same vertical plane as the telescope
- leveling head – supports the telescope and permits the bubble in the tube to be centered by means of the leveling screws

2. Wye Level

- has a detachable telescope which rests in supports called wyes
- curved clips – used to fasten the telescope in place

3. Builder's Level

- also called a construction level or architect's level
- used primarily in building construction where a high degree of precision is not a primary requisite
- horizontal circle – used when measuring or laying out horizontal angles

4. Automatic Level

- does not use a level vial and its ability to level itself depends upon the action of a complex pendulum-and-prism device
- equipped with a prismatic device called a compensator which is suspended on fine, non-magnetic wires

5. Tilting Level

- can be tilted or rotated about its horizontal axis
- tilting knob – used to rotate the telescope into a correct horizontal position
- employed for very precise leveling operations and is equipped with a horizontal circle which makes it suitable for layout and construction surveys

6. Geodetic Level

- a level instrument where most of its metal parts are made of invar to reduce the effects of temperature
- employed in first-order leveling work where extreme precision is an important requirement
- equipped with stadia hairs and vertical and horizontal cross hairs which makes it suitable for three-wire leveling

7. Transit as a Level

- engineer's transit – universal surveying instrument

8. Laser Level

- usually attached to conventional surveying instruments and uses laser light for leveling work

9. Hand Level

- hand-held instrument used on surveys involving short sights
- consists of a brass tube about 15 cm long having a plain glass objective and a peep sight eye-piece

LEVELING ROD

- a graduated rod used for measuring the vertical distance between the line of sight through a leveling instrument and the point whose elevation is either required or known.
- may either be:
 - **Self-Reading Rod** – it can be read directly by the instrument man through the telescope by noting the apparent intersection of the horizontal hair on the rod

- **Target Rod** – has a sliding target which is set and read by a rodman at the position selected by the instrument man (used when longer distances are involved)

OTHER TYPES OF ROD

1. Rods Named After Cities or States

- ex: Philadelphia, Detroit, Chicago, Florida, Boston, New York, Troy, and San Francisco rods
- Philadelphia Rod – combination of self-reading and target rod. It is made in two sections in which the rear section slides over the front section.
- Chicago Rod – comes in three sliding sections and usually extends to either three or four meters. It is graduated similar to the Philadelphia rod except that the figures on the face of the rod are wider and is suitable for longer distances.

2. Rod Ribbons

- an improvised type of rod used in leveling work.
- the graduations on this rod are marked either on canvass or metal strips which are attached to a long piece of selected lumber by staples
- can be easily removed from the wood to which it is attached, rolled, and put into one's pocket after usage

3. Precise Rod

- a form of rod ribbon which uses a graduated invar strip permanently fastened to a 4-meter long wooden or metal frame

4. Geodetic Rod

- a form of rod ribbon which uses a graduated invar metal strip.
- the graduations on this rod are painted upside down for use with inverting telescopes

5. Tape Rod

- also known as the automatic rod
- used when numerous elevations are to be determined from a single set-up of the leveling instrument
- a 3-meter long graduated metal tape is looped around the frame of the rod by means of rollers located at both ends of the frame

ROD LEVEL

- a device used for fast and correct plumbing of a rod
- L-shape in design
- consists of a small circular spirit level fastened to the rod or to a small bracket held against the side of the rod

TARGET

- a small device attached to a rod when extremely long sights make direct reading of the rod difficult or impossible

TELESCOPE

- a metal tube containing a system of lenses which are used to fix the direction of the line of sight and in magnifying the apparent size of objects in its field of view
- the instrument was suggested by Johannes Kepler to be employed for use in surveying
- **external focusing telescope** – the objective lens is mounted on a sleeve which moves back and forth in the telescope barrel as an object is brought into focus
- **internal focusing telescope** – have an additional auxiliary lens which moves back and forth between the objective and the cross hairs as the focusing screw is turned

PARTS OF A TELESCOPE

1. Objective Lens

- a compound lens composed of crown and flint glass mounted in the objective end of the telescope and has its optical axis concentric with the tube axis
- allows light rays to enter the telescope and form an image of the object sighted within its field of view

2. Eyepiece

- a form of microscope containing either 2 or 4 lenses used to enlarge altogether the image and the cross hairs
- allows the instrument man to sight and read accurately the graduations on a leveling rod
- **erecting eyepiece** – consists of 4 lenses which both magnify and erect the image

3. Cross Hairs

- consists of a pair of lines which are perpendicular to each other and are used to define the instrument's line of sight

MAGNIFICATION

- ratio of the apparent size of an object viewed through a telescope to its size as seen by the unaided eye from the same distance (expressed in diameters)

LEVEL VIAL

- a sealed graduated glass tube containing some amount of liquid and a small air bubble

COINCIDENCE BUBBLE

- a type of bubble used on most modern and precise instruments such as the tilting and automatic levels
- employs an optical device which splits the bubble longitudinally then turns one end around to make it appear adjacent to the other end

TRIPOD

- serves as a base to prevent movement of the instrument after it is set up
- consists of three wooden or aluminium legs which are securely fastened to the tripod head by means of a hinged point
- **fixed-leg tripod** – a tripod whose legs are made of solid wood
- **extension tripod** – a tripod whose legs have a sliding section

SHOE

- a pointed piece of metal attached at the end of a tripod
- the tripod is forced into the ground by stepping on it with one's foot

SOURCES OF ERROR IN LEVELING

1. Instrumental Errors

- Instrument out of adjustment
- Rod not of standard length
- Defective tripod

2. Personal Errors

- Bubble not centered
- Parallax
- Faulty rod readings
- Rod not held plumb
- Incorrect setting of target
- Unequal backsight and foresight distances

3. Natural Errors

- Curvature of the earth
- Atmospheric refraction
- Temperature variations

- Wind
- Settlement of the instrument
- Faulty turning points

COMMON MISTAKES IN LEVELING

- Misreading the rod
- Incorrect recording
- Erroneous computations
- Rod not fully extended
- Moving turning points

* **Two-peg test** – a procedure done to check and adjust the line of sight

CURVATURE AND REFRACTION

- Due to **earth's curvature**, a horizontal line departs from a level line by 0.0785 m in 1 km, varying as the square of the length of the line
- Due to **atmospheric refraction**, a horizontal line departs from a level line by 0.0110 m in 1 km, varying as the square of the length of the line
- Due to the **combination of the earth's curvature and atmospheric refraction**, the line of sight varies from a level line by approximately 0.0675 m in 1 km

$$h' = 0.0675 K^2 \quad (K \text{ is in km})$$

DIFFERENTIAL LEVELING

- process of determining the difference in elevation between two or more points some distance apart.
- requires a series of set ups of the instrument along a general route and, for each set up, a rod reading back to the point of known elevation and forward to a point of unknown elevation are taken

DEFINITION OF TERMS

1. Bench Mark

- a fixed point of reference whose elevation is either known or assumed
- **permanent bench marks (PBM)** – established at intervals throughout the country by the Philippine Coast and Geodetic Surveys (PCGS) or the Bureau of Lands. It consists of bronze or brass disks which are permanently set in concrete foundations. They are marked with the elevation above sea level, the year established, and its reference number.

- **temporary benchmarks (TBM)** – those set up by the surveyor for his own use in a particular surveying project and may have assumed elevations

2. Backsight

- a reading taken on a rod held on a point of known or assumed elevation
- referred to as plus sights (+S) as they are added to the elevation of points being sighted to determine the height of instrument

3. Foresight

- a reading taken on a rod held on a point whose elevation is to be determined
- referred to as minus sights (-S)

4. Backsight Distance (BD)

- horizontal distance from the center of the instrument to the rod on which a backsight is taken

5. Foresight Distance (FD)

- horizontal distance from the center of the instrument to the rod on which a foresight is taken

6. Turning Point (TP)

- an intervening point between two bench marks upon which point foresight and backsight rod readings are taken to enable a leveling operation to continue from a new instrument position
- also referred to as change point (CP)

7. Height of Instrument

- also called height of collimation
- the elevation of the line of sight of an instrument above or below a selected reference datum

DOUBLE-RODDED DIFFERENTIAL LEVELING

- a method of determining differences in elevation between points by employing two level routes simultaneously

THREE-WIRE LEVELING

- a more precise method of differential leveling wherein three horizontal hairs (or threads) are read and recorded rather than a single horizontal hair

PROFILE LEVELING

1. Profile

- curved line which graphically portrays the intersection of a vertical plane with the surface of the earth

2. Stationing

- a numerical designation given in terms of horizontal distance any point along a profile line is away from the starting point

3. Intermediate Foresights

- also known as ground rod readings which are taken along the centreline of the proposed project to provide an accurate representation of the ground surface
- sights observed at regular intervals (10 to 30 meters along the centerline) and at points where sudden changes in elevation occur

4. Full Stations

- points which are established along the profile level route at uniformly measured distances

5. Plus Stations

- any other intermediate point established along a profile level which is not designated as a full station

6. Vertical Exaggeration

- process of drawing the vertical scale for a profile much larger than the horizontal scale in order to accentuate the differences in elevation

7. Profile Paper

- a special heavy grade graphing paper used for plotting profiles

Information Found in the Profile Paper:

- Plotting Scales
- Stationing
- Reference Points
- Marginal Information

RISE AND FALL METHOD OF LEVELING

- based on the principle that two consecutive readings from the same instrument position gives the difference in elevation of the two points sighted

RECIPROCAL LEVELING

$$DE_1 = (a - b)$$

$$DE_2 = (a' - b')$$

$$TDE = \frac{(DE_1 + DE_2)}{2} = \frac{(a - b) + (a' - b')}{2}$$

TRIGONOMETRIC LEVELING

- **Difference in elevation between A and B:**

$$DE_{ab} = d \tan \alpha + HI - RR$$

$$DE_{ab} = s \sin \alpha + HI - RR$$

- **For horizontal distances exceeding 300 meters, accounting the effects of curvature and refraction:**

$$DE_{ab} = d \tan \alpha + HI - RR \pm 0.0675K^2$$

$$DE_{ab} = s \sin \alpha + HI - RR \pm 0.0675K^2$$

* effect of curvature and refraction:

added – upward sight

subtracted – downward sight

- **For reciprocal vertical-angle observation:**

$$DE_{ab} = \frac{d}{2} (\tan \alpha + \tan \beta)$$

$$DE_{ab} = \frac{s}{2} (\sin \alpha + \sin \beta)$$

BAROMETRIC LEVELING

- ideal in determining large differences in elevation in rough or mountainous terrain where extensive areas need to be covered
- results are better obtained when the weather is stable and climatic conditions are constant
- such surveys are best undertaken at about mid-morning or mid-afternoon (readings at about noon time, early in the morning, or late in the afternoon should be avoided)

BAROMETER

- an instrument for measuring variations in atmospheric pressure
- first barometer → devised by Evangelista Torricelli
 - **Mercurial Barometer**
 - measures atmospheric pressure by the height of a column of mercury supported by the atmosphere in an evacuated glass tube
 - under standard conditions, the mercury column is about 76 cm high at sea level
 - a centimeter in the height of the mercury column corresponds to a difference of about 108 m in altitude
 - **standard atmospheric pressure at sea level:** 1013.2 millibars

- **Aneroid Barometer**

- invented by Lucien Vidie of France
- functions by using an airtight metal box which responds to changes in air pressure

- **Altimeter**

- precise aneroid barometers designed specifically for surveying applications

ALTIMETER SURVEYS

1. Single-Base Method

- in this method, two altimeters and two thermometers are employed
- one altimeter remains at a reference base whose elevation is known while the other altimeter (roving altimeter) is taken to other points or field stations whose elevations are to be determined

2. Two-Base Method

- designed to eliminate the need to apply correctness for the effects of temperature and relative humidity
- one base is established at a suitable low point (lower base) while a second base is positioned at a much higher elevation (upper base)

$$\frac{S'L'}{SL} = \frac{HL'}{HL}$$

INVERSE LEVELING

- done to determine the elevation of a point located higher than the telescope of the instrument
- in this situation, the rod is held upside down and its base is placed up at the desired point

CROSS SECTION LEVELING

1. Roadway Cross-Sections

- a type of cross-section required for most route projects such as roads and railroads
- planimeter – used in determining the cross-section areas

2. Borrow-Pit Cross-Sections

- borrow pit – an open area which is usually adjacent to a construction project where suitable fill material is excavated

LEVEL CIRCUIT

- a line of levels which begins and ends at the same point or is run between points of precisely known elevations

$$E_c = \pm(Elev_{observed} - Elev_{known})$$

$$Corr = -\left(\frac{E_c}{D}\right)d$$

FOUR TYPES OF MERIDIAN

1. True Meridian

- also known as astronomic or geographic meridian
- this line passes through the geographic north and south poles of the earth and the observer's position

2. Magnetic Meridian

- a fixed line of reference which lies parallel with the magnetic lines of force of the earth

3. Grid Meridian

- a fixed line of reference parallel to the central meridian of a system of plane rectangular coordinates

4. Assumed Meridian

- an arbitrary chosen fixed line of reference which is taken for convenience
- this meridian is usually the direction from a survey station to an adjoining station or some well-defined and permanent point

EXPEDIENT METHODS OF ESTABLISHING MERIDIANS

1. Establishing Magnetic Meridian by Compass

2. Determining True North by Aid of Sun and a Plumb

3. Determining True North by the Rising and Setting of the Sun

4. Determining True North by Polaris

- * Big Dipper – a useful reference constellation of the northern hemisphere
- * The two stars, Merak and Dubhe, forming the side of the dipper which is farthest from the handle are known as the pointer stars

5. Determining True South by the Southern Cross

- * Southern Cross – a constellation of the southern hemisphere which serves as a reference group of stars for determining the location of the earth's south pole

6. Determining Direction of True North (or South) by a Wrist Watch

DESIGNATION OF NORTH POINTS

1. True North

- the north point of the true meridian
- symbol: asterisk or TN

2. Magnetic North

- a north point that is established by means of a magnetized compass needle when there are no local attractions affecting it
- symbol: half arrowhead or MN

3. Grid North

- a north point which is established by lines on a map which are parallel to a selected central meridian
- symbol: full arrowhead or GN or Y

4. Assumed North

- used to portray the location of any arbitrarily chosen north point
- symbol: small black circle or AN

DIRECTION OF LINES

1. Interior Angle

- angle between adjacent lines in a closed polygon
- re-entrant angle – an interior angle that is greater than 180°

2. Deflection Angle

- angle between a line and the prolongation of the preceding line

3. Angle to the Right

- angles that are measured clockwise from the preceding line to the succeeding line

4. Bearing

- the acute horizontal angle between the reference meridian and the line
- forward bearing – when the bearing of a line is observed in the direction in which the survey progresses
- back bearing – when the bearing of the line is observed in an opposite direction

5. Azimuth

- angle between the meridian and the line measured in a clockwise direction from either the north or south branch of the meridian

COMPASS

- a hand-held instrument for determining the horizontal direction of a line with reference to the magnetic meridian

ESSENTIAL FEATURES OF THE MAGNETIC COMPASS

1. **Compass Box**
2. **Line of Sight**
3. **Magnetic Needle**

TYPES OF COMPASS

1. Brunton Compass

- combines the main features of a prismatic compass, sighting compass, hand level, and clinometer
- consists of a brass case hinged on two sides

2. Lensatic Compass

- consists of an aluminum case containing a magnetic dial balanced on a pivot, a hinged cover with a sighting wire, a hinged eyepiece containing a magnifying lens for reading the dial graduations, and a sighting slot for viewing the distant object
- designed for military use

3. Surveyor's Compass

- its main parts include a compass box containing a graduated circle, two sight vanes, a magnetic needle, and two clamping screws

4. Plain Pocket Compass

- similar to surveyor's compass except that it has no sight vanes

5. Prismatic Compass

- the graduations are found on a rotating card instead of being on the compass box

6. Forester's Compass

- a type of pocket compass which is usually made of aluminum or some type of metal which does not affect the free movement and positioning of the magnetic needle

7. Transit Compass

- has a compass box similar to surveyor's compass which is mounted on the upper plate of the transit and often used to check horizontal angles and directions measured or laid off during transit surveys

MAGNETIC DECLINATION

- the horizontal angle and direction by which the needle of a compass deflects from the true meridian at any particular locality

VARIATIONS IN MAGNETIC DECLINATION

1. Daily Variation

- also called diurnal variation
- an oscillation of the compass needle through a cycle from its mean position over a 24-hour period
- extreme eastern position of the needle → occurs early in the morning
- extreme western position of the needle → occurs just about after noon time
- daily variation is greater in higher latitudes than near the equator

2. Annual Variation

- another form of periodic swing taken by the magnetic meridian with respect to the true meridian
- it usually amounts to only less than 1 minute of arc

3. Secular Variation

- covers a period of so many years that its exact cause and character is not thoroughly understood

4. Irregular Variation

- a type of variation uncertain in character and cannot be predicted as to amount or occurrence

ISOGONIC CHART

- a chart or map which shows lines connecting points where the magnetic declination of the compass needle is the same at a given time
- **agonic lines** – lines connecting parts of the chart with zero magnetic declination

* In areas west of the agonic line, the needle has an easterly declination

* In areas east of the agonic line, the needle has a westerly declination

LOCAL ATTRACTION

- any deviation of the magnetic needle of a compass from its normal pointing towards magnetic north

MAGNETIC DIP

- a characteristic phenomenon of the compass needle to be attracted downward from the horizontal plane due to the earth's magnetic lines of force

COMPASS SURVEYS

1. Traverse

- a series of lines connecting successive points whose lengths and directions have been determined from field measurements

2. Traversing

- process of measuring the lengths and directions of the lines of the traverse for the purpose of locating the position of certain points

3. Traverse Station

- any temporary or permanent point of reference over which the instrument is set up
- sometimes called angle points because an angle is usually measured at such stations

4. Traverse Lines

- lines connecting traverse stations and whose lengths and directions are determined

TYPES OF COMPASS SURVEYS

1. Open Compass Traverse

- consists of a series of lines of known lengths and magnetic bearings which are continuous but do not return to the starting point or close upon a point of known position

2. Closed Compass Traverse

- consists of a series of lines of known lengths and magnetic bearings which forms a closed loop, or begin and end at points whose positions have been fixed by other surveys of higher position

SOURCES OF ERROR IN COMPASS WORK

1. Bent needle
2. Bent pivot
3. Sluggish needle
4. Plane of sight not vertical
5. Electrically charged compass box
6. Local attraction
7. Magnetic variation
8. Errors in reading the needle

THE ENGINEER'S TRANSIT

- invented by Roemer, a Danish astronomer
- consists of a telescope and two large protractors: one mounted in the horizontal plane and the other in the vertical plane
- referred to as the universal surveying instrument

MAIN PARTS OF THE TRANSIT

1. Upper Plate

- also called the alidade
- consists of the entire top of the transit
- rotates about the vertical axis
- **Parts of the Upper Plate:**
 - **Telescope** – used for fixing the direction of the line of sight, viewing the objects, and magnifying their apparent size in the field of view
 - **Standards** – used to hold into position the horizontal axle level so that the telescope can be elevated or depressed by rotating on an axis perpendicular to the line of sight
 - **Compass Box** – for establishing the magnetic meridian and to allow rough checks on measured angles
 - **Plate Level Vials** – used to establish the upper and lower plates in a horizontal plane
 - **Vertical Circle** – used in measuring vertical angles
 - **Plate Verniers** – provided for the horizontal circle
 - **Optical Plummet** – a small telescope through the vertical center of the transit
 - **Telescope Clamp** – used to hold the telescope horizontally or at any desired inclination
 - **Telescope Tangent Screw** – allows the telescope to be rotated slightly or in small movements about the horizontal axis after the telescope is tightened
 - **Upper Clamp** – a locking device attached to the upper plate and rotates with it
 - **Upper Tangent Screw** – allows a small range of movement between the upper and lower plates even after the upper clamp is tightened

2. Lower Plate

- the scale with which horizontal angles are measured
- the underside of the lower plate is attached to a vertical and tapering spindle called the outer spindle
- **Parts of the Lower Plate:**
 - **Lower Clamp** – used to control the rotation of the horizontal circle

- **Lower Tangent Screw** – used to make precise settings after the lower clamp is tightened

3. Leveling Head Assembly

- allows the transit to be leveled and centered over a point
- **Parts of the Leveling Head Assembly:**
 - **Leveling Screws** – used for leveling the instrument by the plate levels
 - **Plumb Bob Chain** – used for attaching a string and a plumb bob so that the instrument may be set exactly over a selected point on the ground

VERNIERS

- small graduated mechanical device attached and made to slide along linear or circular scales in order to increase to a higher degree of accuracy the readings obtained on such scales
- invented by Pierre Vernier

TWO TYPES OF VERNIERS

1. Direct Vernier

- a type of vernier containing n-1 divisions, and divided into n divisions on the vernier scale

2. Retrograde Vernier

- a type of vernier containing n+1 divisions, and divided into n divisions on the vernier scale

$$LC = \frac{s}{n}$$

where:

LC – least count (smallest division that can be read on the main scale)

s – value of the smallest space or division on the main scale

n – number of divisions on the vernier

* There are 3 verniers in an engineering transit. The two are used on the horizontal circle, and one at the vertical circle

THREE TYPES OF VERNIERS USED IN TRANSITS

1. Single Vernier

- vernier used for the measurement of angles where a high degree of precision is required such as triangulation work
- it is read in only one direction and must be set with the graduations ahead of the zero mark in the direction to be turned
- least count: 10 seconds

2. Double Vernier

- consists of two parts: vernier on the left (for reading clockwise circle graduations) and vernier on the right (for reading counterclockwise circle graduations)
- least count: 30 seconds

3. Folded Vernier

- it has a length similar to a single vernier, however, half of the graduations are placed on each side of the index mark
- used when there is not enough space available for a double vernier
- least count: 1 minute

CLOSING THE HORIZON

- process of measuring horizontal angles about a point
- **horizon misclosure** – difference between 360 degrees and the sum of the measured angles

MEASURING VERTICAL ANGLES

$$\alpha' = \frac{(\alpha_N + \alpha_R)}{2}$$

where:

α' – correct value of the measured vertical angle

α_N – vertical angle measured with telescope in direct or normal position

α_R – same vertical angle measured with telescope in reversed or plunged position

INDEX CORRECTION

- **index error** – error introduced when observing a vertical angle when the line of sight is not parallel to the axis of the telescope level tube resulting to an inclination of the vertical axis

$$IE = \frac{(\alpha_N - \alpha_R)}{2}$$

METHODS IN PROLONGING A STRAIGHT LINE

1. Double Centering

2. Balancing In

ANGLES BY REPETITION

$$\alpha_{ave} = \frac{RDG_{last} + No. of Passes (360^\circ) - RDG_{initial}}{n}$$

ADJUSTMENT OF THE TRANSIT

- **Shop adjustment** – adjustment made in a service facility or the manufacturer of the instrument or through an authorized dealer or distributor
- **Field adjustment** – undertaken by the instrument man periodically or just prior to executing a survey

THEODOLITE

- an instrument designed to accomplish the same primary purpose as a transit, that is to measure horizontal and vertical angles and to prolong straight lines

TYPES OF THEODOLITES

1. Repeating Theodolite

- designed so that it can measure a horizontal angle as many times as required by adding them successively on the graduated circle

2. Directional Theodolite

- the horizontal circle remains fixed during a series of observations

3. Digital Theodolite

TRAVERSE COMPUTATIONS

• Latitude

- the projection of a line onto the reference meridian or a north-south line

• Departure

- the projection of a line onto the reference parallel or an east-west line

$$Lat = d \cos \beta$$

$$Dep = d \sin \beta$$

$$\tan \beta = \left(\frac{-\sum Dep}{\sum Lat} \right)$$

$$LEC = \sqrt{(\sum Lat)^2 + (\sum Dep)^2}$$

$$REC = \frac{LEC}{D}$$

TRAVERSE ADJUSTMENT

- procedure of computing the linear error of closure and applying corrections to the individual latitudes and departures for the purpose of providing a mathematically closed figure

COMPASS RULE

- named after Nathaniel Bowditch
- it is based on the assumption that all lengths were measured with equal care and all angles taken with approximately the same precision

$$c_l = c_L \left(\frac{d}{D} \right)$$

$$c_d = c_D \left(\frac{d}{D} \right)$$

TRANSIT RULE

- rule in traverse adjustment where the latitude and departure corrections depend on the length of the latitude and departure of the course respectively instead of both depending on the length of the course

$$c_l = \frac{c_L |Lat|}{\sum |Lat|}$$

$$c_d = \frac{c_D |Dep|}{\sum |Dep|}$$

CRANDALL METHOD

- method of adjusting a traverse by applying the theory of least squares
- it is suitable for use when the linear measurements made are less precise than the angular measurements

TRAPEZOIDAL RULE

$$A = \left(\frac{h_1 + h_2}{2} \right) d + \left(\frac{h_2 + h_3}{2} \right) d + \dots + \left(\frac{h_{n-1} + h_n}{2} \right) d$$

SIMPSON'S ONE-THIRD RULE

$$A = \frac{d}{3} [(h_1 + h_n) + 2(odd) + 4(even)]$$

PLANIMETER

- a mechanical device used for determining the area of any shape of figure plotted to a known scale